EXPERIMENT 2

EXPT NO: 2 OCC AND LOAD CHARACTERISTICS OF D.C SHUNT GENERATOR

Aim:

To conduct load tests on separately excited generators and to obtain the characteristics.

Apparatus Required:

Sl .no	Apparatus	Range	Type	Quantity
1	Voltmeter	(0-300)V	MC	1
2	Ammeter	(0-2.5)A	MC	1
3	Ammeter	(0-5)A	MC	1
4	Rheostat Wire	-	wound	1
5	Rheostat Wire	-	wound	1
6	Connecting wires	-	-	As required

Fuse rating calculation for field and armature:

No load test

10 % of rated current (full load current)

Load test

125 % of rated current (full load current)

Precautions:

1. Motor side field rheostat should be kept at minimum resistance position.
2. Generator side field rheostat should be kept at maximum resistance position.
3. The starter should be in an off position before switching on the supply.
4. The DPST switch must be kept open.Procedure for open circuit test Procedure
5. Connections are given as per the circuit diagram.
6. The motor is started with the help of a THREE POINT starter.
7. Adjust the motor speed to rated speed by adjusting motor field rheostat when the generator is disconnected from the load by DPST switch 2.

8. By varying the generator field rheostat gradually, the open circuit voltage [Eo] and corresponding field current (If) are tabulated up to 125 % of rated voltage of the generator.
9. The motor is switched off by using DPST switch 1 after bringing all the rheostats to initial position.

Procedure for Load test:

1.Make the connections as per the instructions.

2.Start the prime mover using the three-point starter and maintain the rated speed

when the generator is disconnected from the load by DPST switch 2.

3.Adjust the generator field rheostat gradually and note the values.

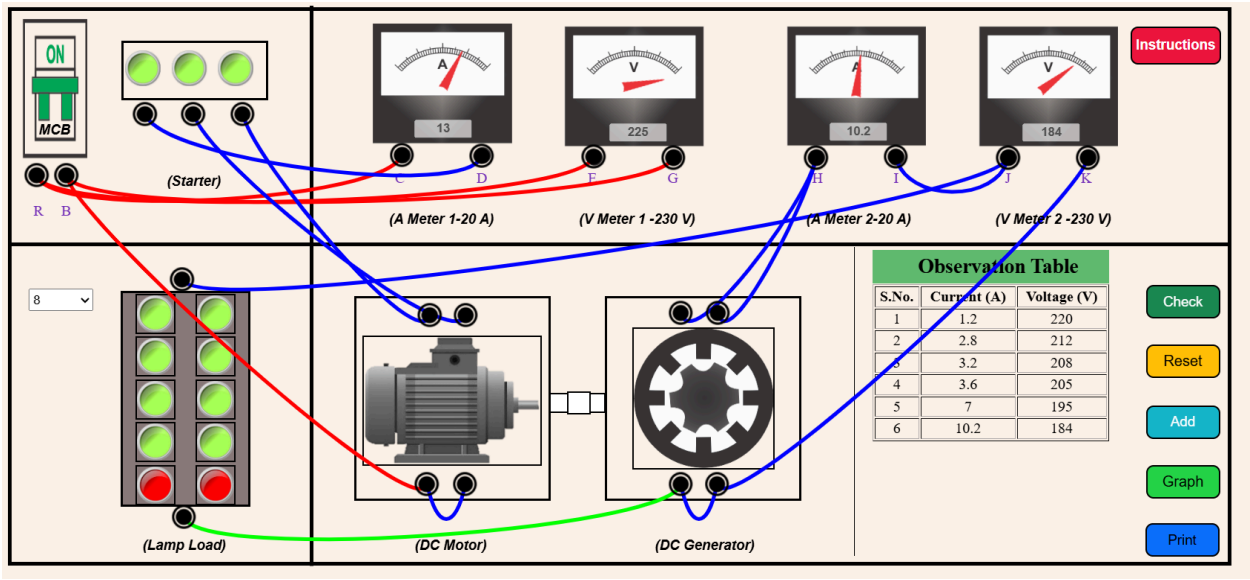
4.Record the ammeter and voltmeter readings under no-load conditions.

5.Connect the load and gradually increase it up to the rated current,noting the readings.

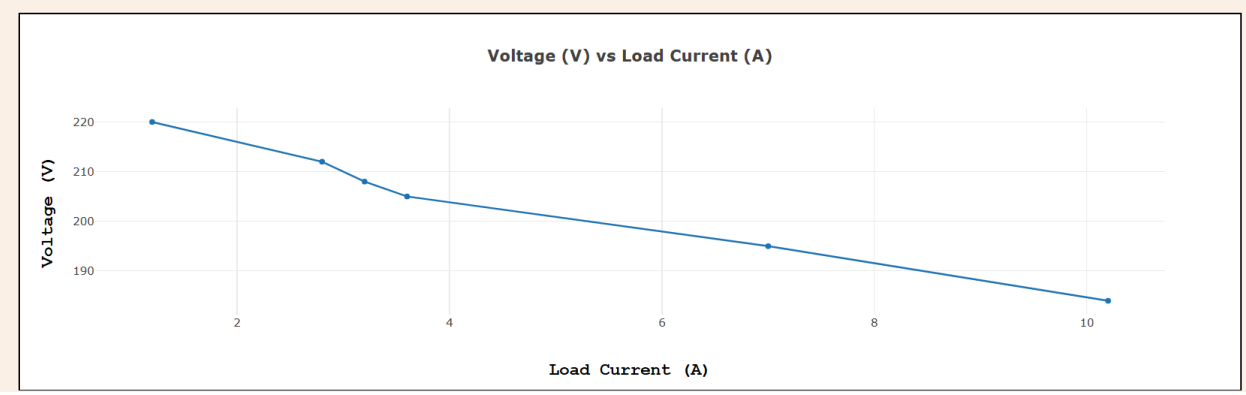
6.Tabulate all the readings for the open-circuit and load tests.

7.Bring the load and rheostats to their initial positions and switch OFF the motor.

Tabulation for Open Circuit Test:



Model graph:



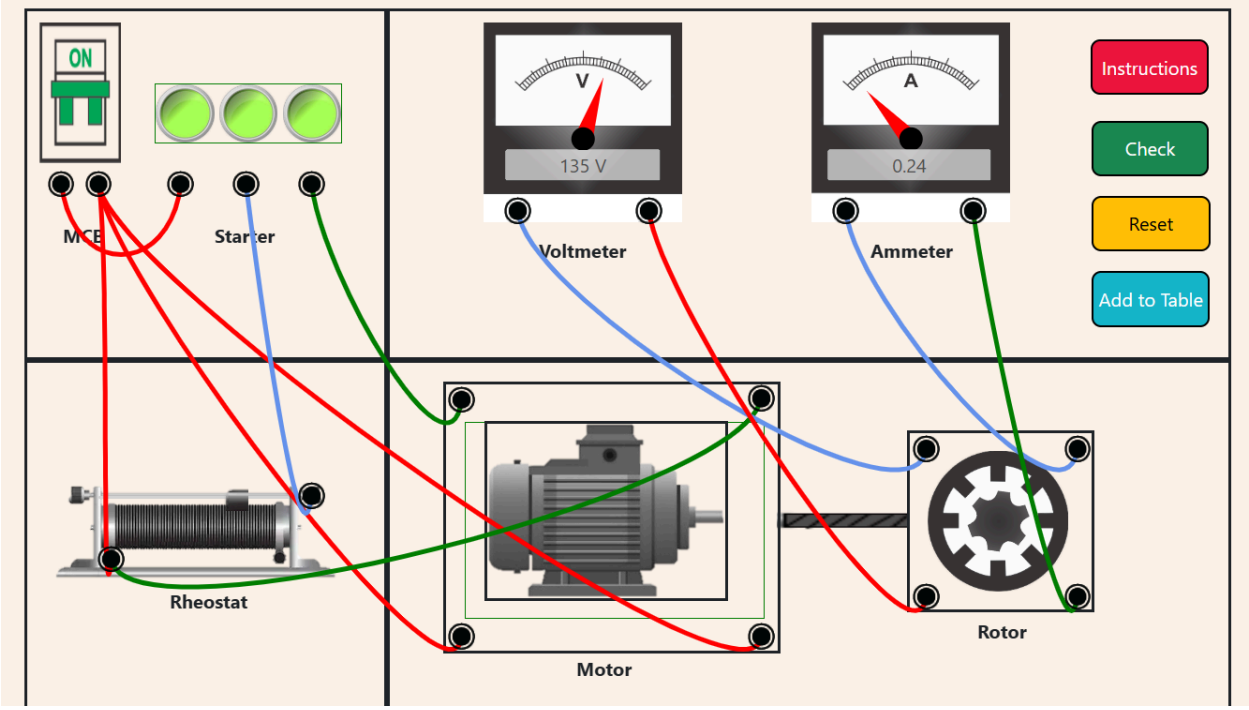
Calculation:

Observation Table

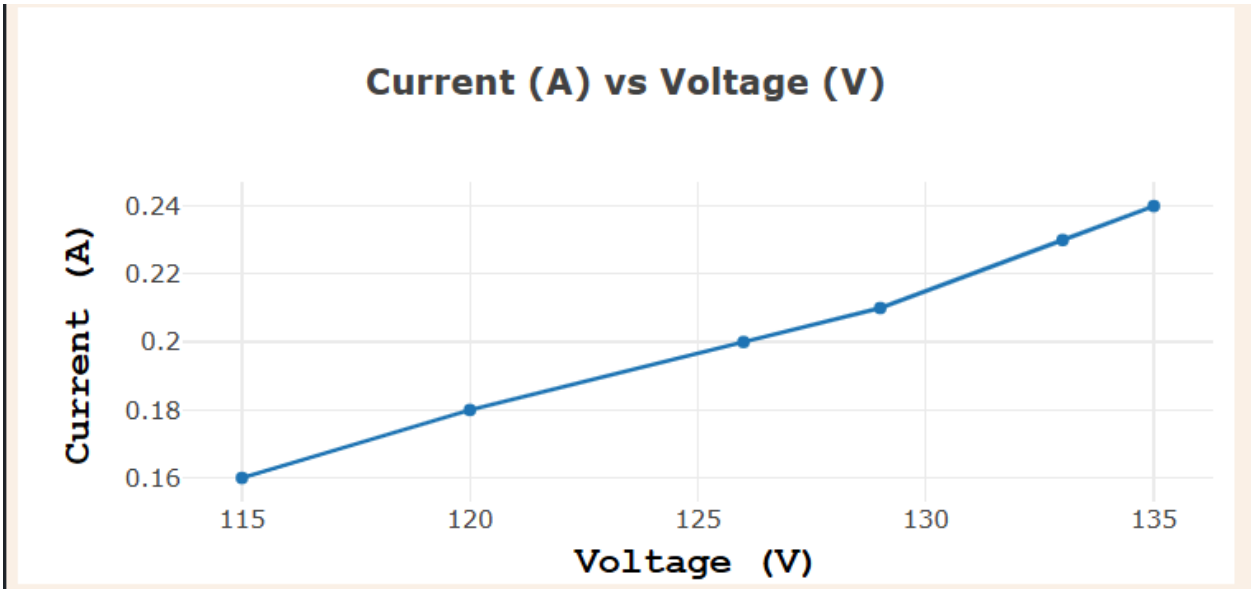
S.No.	Current (A)	Voltage (V)
1	1.2	220
2	2.8	212
3	3.2	208
4	3.6	205
5	7	195
6	10.2	184

Magnatization Load Characteristics:

Tabulation for Load Test:



Model Graph:



Calculation:

OBSERVATION TABLE

S.No.	Voltage (V)	Current (A)
1	115	0.16
2	120	0.18
3	126	0.20
4	129	0.21
5	133	0.23
6	135	0.24

Result:

The load test on separately excited generators and to obtain the characteristics was verified.